

(3) Every eigenvalue of orthogonal matrix has absolute value 1. (8 pts)

True.

Let $u = \frac{x}{\|x\|}$,

$$Au = \lambda u$$

$$A \cdot \frac{x}{\|x\|} = \lambda \frac{x}{\|x\|}$$

$$A \cdot \frac{1}{\|x\|} = \lambda$$

請先把 orthogonal 的定義 ()
matrix
↑ 先寫一個

$$A^{-1} = A^T$$

再寫其性質 (Thm 7.1.3)

$$\langle Ax, Ax \rangle = x^T A^T A x = x^T x$$

再寫 eigenvalue 的定義
你就會得到 $Ax = \lambda x$.

$$Ax = \lambda x \text{ for } x \neq 0$$

$$\langle Ax, Ax \rangle = \langle \lambda x, \lambda x \rangle$$

$$= \lambda^2 \langle x, x \rangle$$

$$= \lambda^2 x^T x$$

2. Determine whether the statement is true or false, and justify your answer.

T (1) If A is diagonalizable then A^{-1} is diagonalizable. (2 pts)

If A is diagonalizable, then there exists an invertible matrix P and a diagonal matrix such that $P^{-1}AP = D$. "Since A is similar to D , A is invertible $\Leftrightarrow D$ is invertible."

$$\textcircled{1} P^{-1}AP = D \Leftrightarrow A = PDP^{-1}, A^{-1} = (PDP^{-1})^{-1} = (P^{-1})^{-1}D^{-1}P^{-1} = PD^{-1}P^{-1}, A^{-1} \text{ is diagonalizable.}$$

$$\textcircled{2} (P^{-1}AP)^{-1} = P^{-1}A^{-1}(P^{-1})^{-1} = P^{-1}A^{-1}P = D^{-1} \quad * \text{ 要將 } (ABC)^{-1} \text{ 折成 } C^{-1}B^{-1}A^{-1} \text{ 之前, 需先確認 } A, B, C \text{ 分別可逆}$$

由題目可知 A^{-1} 存在, 由對角化定義可知 P^{-1} 存在, 若從 $\textcircled{1}$ 推導則需說明 D^{-1} 存在, 從 $\textcircled{2}$ 則不需要

T (2) Every orthogonal matrix is invertible. (2 pts)

By the definition of the orthogonal matrix,

$$A^T A = A A^T = I \Leftrightarrow A^{-1} = A^T,$$

Thus, every orthogonal matrix is invertible.

T (3) Every eigenvalue of orthogonal matrix has absolute value 1. (8 pts)

Let A be an orthogonal matrix

λ be an eigenvalue of A

x be an eigenvector of A w.r.t. λ , $x \neq 0$

$$Ax = \lambda x$$

$$\begin{aligned} (Ax) \cdot (Ax) &= x^T A^T A x \\ &= x^T (A^T A) x \\ &= x^T x \\ &= \lambda^* x^T x \\ &= (\lambda x) \cdot (\lambda x) \end{aligned}$$

$$\Rightarrow x^T x = \|x\|^2 = \lambda^* \|x\|^2 = \lambda^* x^T x \quad \text{因 } x \neq 0, \text{ 才可相消}$$

$$\therefore \|x\| \neq 0$$

$$\therefore \lambda^* = 1 \Rightarrow |\lambda| = 1$$